

DER-CHIEN CHANG

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LinkedIn | GitHub | AI Blog

Education

University of Toronto	2026
Master of Computer Engineering – Data Analysis & Machine Learning	
University of British Columbia	2024
Bachelor of Applied Science – Electrical Engineering	

Work Experience

AI Intern, Ret[AI]ling Data JUNE 2025 – DEC 2025

- Migrated a tightly coupled, hand-engineered RAG pipeline to a modular LangGraph agent architecture, introducing conversational agent based architecture and standardized database vector search, enabling rapid chatbot deployment via data configuration and domain specific system prompt
- Built a minimum viable product (MVP) multi-agent AI system capable of generating interacting graph from the database based on user's natural language input.
- Designed a high-precision RAG pipeline using MongoDB Atlas Vector Search + OpenAI embeddings with hybrid retrieval and adaptive re-ranking, achieving 90–93% retrieval precision on 2,000+ QA pairs and tracking quality with RAGAS metrics.
- Implemented a scalable, containerized FastAPI backend for a production chatbot, using Redis for session-based conversation state and MongoDB for per-user data persistence, supporting 1,000+ daily users with end-to-end latency less than 300 ms.
- Implemented RBAC for a third-party messaging chatbot, combining email verification with role enforcement to prevent unauthorized access and reduce the risk of sensitive data leakage in multi-tenant deployments.

Software Intern, Anyload Weigh & Measure SEPT 2022 – DEC 2022

- Built automated Python ETL pipelines and relational database schemas to keep inventory data synchronized across warehouses, improving data reliability for real-time stock tracking and operational reporting.

Technical Projects

NLP Fine-Tuning & Training Platform Jan 2026 – Mar 2026

Skills: PyTorch, Hugging Face Transformers, FastAPI, Redis, S3 (boto3), Docker, GitHub Actions

- Built an end-to-end NLP pipeline for text classification with configurable preprocessing and fine-tuning of transformer backbones (BERT, DistilBERT, RoBERTa, Longformer), supporting frozen/partial/full unfreeze and modular classifier heads (MLP, BiGRU).
- Engineered a FastAPI service to orchestrate asynchronous training and inference via background jobs, with real-time progress tracking, session-based experiment control, and scalable RESTful model serving.
- Designed a modular ML system with Redis-backed job/state management and S3 checkpointing to support concurrent training workflows; implemented Docker + GitHub Actions CI/CD for versioned deployment to GPU environments.

Airport Analytics Agent Aug 2025 – Dec 2025

Skills: LangGraph, LangChain, FastAPI, Redis, AWS S3, Docker

- Architected a production-ready multi-agent analytics system using LangGraph's supervisor pattern, coordinating three specialized agents (SQL search, visualization, analysis) to process conversational queries over 30,000+ airport records.
- Built a visualization pipeline, converting natural-language queries into executable SQL, exporting results to AWS S3 bucket, and dynamically generating interactive Plotly charts with intelligent chart-type selection.
- Deployed a conversational chatbot with Redis persistence, Docker containerization, FastAPI supported backend, and a real-time anomaly detection module that triggered automated alerts for abnormal flight patterns.

MCP: Natural Language to Data Visualization Agent Jul 2025 - Aug 2025

Skills: Model Context Protocol (MCP), LangChain, LangGraph, Docker, SQLite, MCP

- Built an end-to-end MCP learning and demo project that progresses from core Model Context Protocol (MCP) primitives to a production-style LangChain + LangGraph agent application, designed to help developers adopt MCP in real systems.
- Integrated a third-party MCP server (Tako) into a production-style ReAct agent workflow, enabling real-time search and dataset visualization via streamable HTTP, with SQLite checkpointing for persistent multi-turn conversations.
- Implemented a Streamlit interface that streams agent execution step-by-step (tool calls and intermediate outputs), improving debuggability and user trust in the system's results.

Technical Skills

- **AI Agent & LLMs:** LangGraph, Semantic Kernel, RAG, LangChain, Prompt Engineering, OpenAI APIs
- **MLOps & Backend:** FastAPI, Docker, Redis, AWS, ETL Processes, Git/GitHub
- **Machine Learning & Data:** PyTorch, Scikit-learn, Vector Search (FAISS), SQL, MongoDB